

Local Softmax and Global Weights in Non-Boolean Event Structures

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Softmax and related normalized response functions are widely used in choice theory, machine learning, and cognitive science. In non-Boolean event structures with overlapping contexts, however, local normalization does not automatically yield a global probability weight. We show that imposing single-valuedness on shared atoms—equivalently, no-disturbance or consistent connectedness—collapses generalized softmax rules to coordinate parametrizations of the strictly positive part of the admissible-weight polytope. Any strictly positive admissible weight can be represented in this way, while boundary weights arise as limits. Exotic weights that exceed classical or quantum bounds are therefore properties of the event structure and the chosen weight, not of the normalizing link. The resulting hierarchy separates local normalization, cross-context gluing, Cauchy–Gleason linearity, and physical or cognitive realizability.

I. INTRODUCTION

The softmax rule

$$P_i = \frac{\exp(\beta u_i)}{\sum_j \exp(\beta u_j)} \quad (1)$$

is one of the most common ways of transforming scores, utilities, energies, or response tendencies into probabilities. It appears in statistical mechanics, discrete-choice theory, machine learning, cognitive modeling, and social-science response models. Its appeal is obvious: it is positive, normalized, differentiable, and controlled by an inverse-temperature or discrimination parameter β .

In an ordinary single choice set, Eq. (1) solves the immediate normalization problem. Given a finite list of alternatives, their probabilities sum to one. In a non-Boolean event structure, however, there is an additional compatibility problem. There may be several overlapping contexts, and the same event may occur in more than one of them. Normalizing separately inside each context does not guarantee that shared events receive the same probability in all contexts in which they occur.

This point is familiar in the language of quantum logics, test spaces, exclusivity graphs, generalized urn models, automata, and partition logics [1–5]. A probability weight on such a structure is not merely a family of normalized distributions, one for each context. It is a single function on atoms that normalizes on every context. If an atom is shared by two contexts, it has one value, not two. This cross-context agreement will be called *single-valuedness on atoms*. It is the finite-test-space form of *no-disturbance*; in Bell-type scenarios the analogous condition is no-signalling, and in the contextuality-by-default literature it is closely related to consistent connectedness.

The purpose of this paper is to clarify the relation between this condition and generalized softmax rules. We

consider normalized response functions of the form

$$P_{g,C}(a) = \frac{g(u_C(a))}{\sum_{b \in C} g(u_C(b))}, \quad (2)$$

where C is a context, $a \in C$ is an atom, $u_C(a)$ is a score, and g is a strictly positive link function. The ordinary softmax is obtained by $g(x) = \exp(\beta x)$. Other choices of g give logistic, probit, polynomial, power-law, or q -exponential response models on suitable domains.

The central observation is elementary but important. If the scores are context-dependent without further constraints, Eq. (2) gives only a family of separately normalized response distributions. Such a family may violate no-disturbance: the same shared atom may receive different probabilities in different contexts. If single-valuedness or no-disturbance is imposed, then the construction becomes exactly a parametrization of an admissible weight on the logic. Conversely, every strictly positive admissible weight can be written in generalized softmax form, provided the link function has a suitable positive range. Thus the softmax wrapper adds no new probability theory once global consistency is required. Its explanatory force begins only when the score functions are themselves constrained by additional modelling assumptions.

This conclusion is useful because the admissible-weight polytope of a non-Boolean event structure can be larger than the classical convex hull of two-valued states. Some logics admit “exotic” weights: normalized, single-valued assignments that are not classical mixtures of dispersion-free states [1, 2]. In odd-cycle scenarios, the admissible weight assigning 1/2 to each cyclic atom and zero to the remaining atoms exceeds the classical bound; for odd cycles of length at least five it also exceeds the usual Lovász/Klyachko–Can–Binicioğlu–Shumovsky (KCBS) quantum value [6–8]. Such a weight is obtained as a limiting softmax assignment. But the beyond-classical or beyond-quantum character belongs to the weight’s location in the polytope, not to the exponential function itself.

There is also a nearby but distinct relation to Cauchy’s

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functional equation. Gleason-type theorems use additivity of probability assignments on sufficiently rich event structures to force linear Born-rule representations; Wright and Weigert have recently made the connection between such Gleason-type arguments and Cauchy’s functional equation explicit [9]. The present result does not derive Born-linearity. It concerns the prior question of when locally normalized softmax responses glue into a single admissible weight. Cauchy’s equation can nevertheless enter again at the level of the link function: if additive score contributions are required to induce multiplicative unnormalized weights, then the exponential link is singled out under the usual regularity assumptions.

The paper is organized as follows. Section II defines finite pasted event structures and their weight polytopes. Section III introduces generalized softmax rules, positive coordinates, and the distinction between local normalization and global gluing. Section IV proves the representation theorem and records the gauge freedom and boundary cases. Section V clarifies the relation to Cauchy’s functional equation and Gleason-type linearity theorems. Section VI relates the construction to classical, quantum, and exotic regions. Section VII treats odd cycles and their half-weights. Section VIII discusses empirical and modelling implications. Section IX collects broader consequences. Section X concludes.

II. PASTED EVENT STRUCTURES AND ADMISSIBLE WEIGHTS

We work with a finite event structure

$$L = (A, \mathcal{M}),$$

where A is a finite set of atoms and $\mathcal{M}$ is a finite family of contexts. Each context $C \in \mathcal{M}$ is a subset of A , interpreted as a maximal set of mutually exclusive and jointly exhaustive outcomes. In a partition-logic realization, the atoms are subsets of a latent state space S , and each context is a partition of S . In an abstract test-space or hypergraph formulation, A and $\mathcal{M}$ are taken as primitive. The event structure is “pasted” in the quantum-logic sense: different Boolean contexts are joined by identifying shared atoms, in analogy with pasting constructions for orthomodular posets and Boolean algebras [10].

We shall reserve the word “gluing” for the probabilistic compatibility condition considered below: locally normalized context-wise distributions glue if they assign the same probability to every atom shared by two or more contexts, thereby defining a single admissible weight on the already pasted event structure.

A probability weight on L is a function

$$p : A \rightarrow [0, 1]$$

satisfying

$$\sum_{a \in C} p(a) = 1 \quad \text{for all } C \in \mathcal{M}. \quad (3)$$

The set of all such weights is a convex polytope:

$$\mathcal{W}(L) = \left\{ p \in [0, 1]^A : \sum_{a \in C} p(a) = 1 \text{ for all } C \in \mathcal{M} \right\}. \quad (4)$$

We call $\mathcal{W}(L)$ the admissible-weight polytope.

An atom $a \in A$ may occur in more than one context. In partition logic such an atom is an intertwining atom: the same subset of the latent state space appears as an atom of two or more partitions. The definition (3) assigns one number $p(a)$ to that atom. Hence single-valuedness on shared atoms is built into the notion of a weight.

A two-valued weight, or dispersion-free state, is a weight

$$v : A \rightarrow \{0, 1\}$$

that assigns value one to exactly one atom in every context. The classical polytope associated with L is the convex hull of all such two-valued weights:

$$\mathcal{C}(L) = \text{conv}\{v : v \text{ is a two-valued weight on } L\}. \quad (5)$$

In a concrete partition-logic model built from a latent state space S , each latent state $s \in S$ induces a point-evaluation state:

$$v_s(B) = \begin{cases} 1, & s \in B, \\ 0, & s \notin B. \end{cases}$$

The convex hull of these intended point states gives the usual classical latent-state model. Abstract logics may possess additional two-valued states, but this distinction is not essential for the softmax question considered here.

A quantum or Born-rule model is obtained when the atoms can be represented by projectors Π_a on a Hilbert space such that

$$\sum_{a \in C} \Pi_a = I \quad \text{for all } C \in \mathcal{M},$$

and probabilities are given by

$$p(a) = \langle \psi | \Pi_a | \psi \rangle.$$

The corresponding quantum set will be denoted schematically by $\mathcal{Q}(L)$. Its precise definition depends on dimension, rank, and representation conventions. For exclusivity-graph formulations, related relaxations are governed by Lovász-theta-type quantities [6, 11].

Whenever a Born-rule representation exists with projectors satisfying context-wise completeness, the associated probabilities define an admissible weight. Indeed,

$$\sum_{a \in C} p(a) = \sum_{a \in C} \langle \psi | \Pi_a | \psi \rangle = \left\langle \psi \left| \sum_{a \in C} \Pi_a \right| \psi \right\rangle = \langle \psi | I | \psi \rangle = 1.$$

Moreover, because the same projector Π_a is assigned to the same atom a , the probability $p(a)$ is single-valued

across contexts. Thus every such Born-rule model lies in $\mathcal{W}(L)$.

The inclusion

$$\mathcal{Q}(L) \subseteq \mathcal{W}(L)$$

follows directly from blockwise completeness of the projectors. If, in addition, one adopts the usual operational convention that classical deterministic assignments are included among the allowed quantum models, then one obtains the familiar chain

$$\mathcal{C}(L) \subseteq \mathcal{Q}(L) \subseteq \mathcal{W}(L).$$

The first inclusion is convention-dependent if one fixes a particular Hilbert-space representation in advance: not every abstract two-valued state need be represented inside that fixed representation. Throughout, claims of being beyond quantum are therefore relative to a specified quantum representation or to a specified graph-theoretic quantum set, such as the Lovász-theta set for exclusivity graphs.

The terminology should also be fixed. In what follows, single-valuedness or no-disturbance means membership in $\mathcal{W}(L)$. Classical noncontextuality, in the stronger sense used in much of the contextuality literature, means membership in $\mathcal{C}(L)$. The distinction is essential: exotic weights are single-valued and normalized on every context, but they need not be classical.

III. GENERALIZED SOFTMAX AND POSITIVE COORDINATES

Let $g : I \rightarrow \mathbb{R}_{>0}$ be a strictly positive link function on an interval $I \subseteq \mathbb{R}$. Given a context $C \in \mathcal{M}$ and scores

$$u_C : C \rightarrow I,$$

define

$$P_{g,C}(a) = \frac{g(u_C(a))}{Z_C}, \quad Z_C = \sum_{b \in C} g(u_C(b)). \quad (6)$$

The ordinary softmax corresponds to $g(x) = \exp(\beta x)$. The choice $g(x) = \Phi(\beta x)$, with Φ the Gaussian cumulative distribution function, gives a probit-type normalized response rule on a suitable range. Logistic, polynomial, power-law, and q -exponential links can be treated similarly, subject to positivity and domain restrictions.

It is useful to factor the construction through the positive coordinates

$$q_C(a) = g(u_C(a)) > 0. \quad (7)$$

Then Eq. (6) becomes simply

$$P_{g,C}(a) = \frac{q_C(a)}{\sum_{b \in C} q_C(b)}. \quad (8)$$

Thus the link function g does not itself carry probabilistic content. It only specifies how positive coordinates $q_C(a)$ are represented by scores $u_C(a)$. The substantive distinction is whether the positive coordinates can be chosen so that the resulting normalized probabilities glue consistently across contexts.

For each fixed context C , Eq. (6) defines an ordinary probability distribution:

$$\sum_{a \in C} P_{g,C}(a) = 1.$$

It also respects additivity for unions of atoms inside that context. Thus generalized softmax solves the *local* normalization problem.

It does not, by itself, solve the *gluing* problem. Suppose that a is an atom contained in two contexts C and C' . In general,

$$P_{g,C}(a) = \frac{q_C(a)}{\sum_{b \in C} q_C(b)}$$

need not equal

$$P_{g,C'}(a) = \frac{q_{C'}(a)}{\sum_{b \in C'} q_{C'}(b)}.$$

The same atom may be assigned two different probabilities. Such a family is a legitimate collection of context-wise response distributions, but it is not a probability weight on L .

We therefore distinguish two notions.

a. Free context-wise generalized softmax. The scores $u_C(a)$, or equivalently the positive coordinates $q_C(a)$, are allowed to depend freely on the context. The model gives a normalized distribution for each C , but no cross-context consistency is assumed.

b. Single-valued generalized softmax. Whenever $a \in C \cap C'$, one requires

$$P_{g,C}(a) = P_{g,C'}(a). \quad (9)$$

Under this condition, the context-wise distributions glue into a single function on atoms.

Condition (9) is the no-disturbance or consistent-connectedness condition for the present finite event structure. It should not be confused with classical noncontextuality. The former gives membership in $\mathcal{W}(L)$; the latter gives membership in $\mathcal{C}(L)$.

The gluing condition can be stated directly in terms of positive coordinates.

Proposition III.1 (Gluing condition for free scores). *Let*

$$q_C(a) = g(u_C(a)) > 0, \quad Z_C = \sum_{b \in C} q_C(b).$$

The context-wise distributions

$$P_C(a) = \frac{q_C(a)}{Z_C}$$

glue to a single weight on atoms if and only if, for every shared atom $a \in C \cap C'$,

$$\frac{q_C(a)}{Z_C} = \frac{q_{C'}(a)}{Z_{C'}}. \quad (10)$$

Equivalently,

$$\frac{Z_C}{Z_{C'}} = \frac{q_C(a)}{q_{C'}(a)} \quad (11)$$

for every $a \in C \cap C'$. In particular, if two contexts share more than one atom, all shared atoms must induce the same normalizer ratio. Around any cycle of overlapping contexts, the product of these ratios must be 1.

Proof. The first statement is just the definition of single-valuedness of the normalized probabilities. Rearranging Eq. (10) gives Eq. (11). If two contexts share several atoms, the same ratio $Z_C/Z_{C'}$ must be obtained from each of them. Multiplying successive ratios around a closed chain of contexts telescopes to 1. $\square$

Proposition III.1 makes explicit why context-independent scores are not sufficient. Even if $q_C(a) = q_{C'}(a)$ for a shared atom, the normalizing denominators Z_C and $Z_{C'}$ may differ. Then the shared atom receives different probabilities.

IV. REPRESENTATION THEOREM

The basic relation between single-valued generalized softmax rules and admissible weights is simple.

Proposition IV.1. *Every single-valued generalized softmax family defines an admissible weight on L .*

Proof. For each atom $a \in A$, choose any context $C \in \mathcal{M}$ such that $a \in C$, and set

$$p(a) = P_{g,C}(a).$$

Condition (9) makes this definition independent of the chosen context. For every $C \in \mathcal{M}$,

$$\sum_{a \in C} p(a) = \sum_{a \in C} P_{g,C}(a) = 1,$$

by the normalization of Eq. (6). Hence $p \in \mathcal{W}(L)$. $\square$

The converse holds under mild assumptions on the link function.

Theorem IV.2 (Representation theorem). *Let $L = (A, \mathcal{M})$ be finite. Let $g : I \rightarrow \mathbb{R}_{>0}$ be strictly monotone on an interval I , and suppose that its range contains an interval $(0, r)$ for some $r > 0$. Then every strictly positive admissible weight $p \in \mathcal{W}(L)$ admits a single-valued generalized softmax representation. That is, there exist scores $u : A \rightarrow I$ such that*

$$p(a) = \frac{g(u(a))}{\sum_{b \in C} g(u(b))} \quad \text{for every } C \in \mathcal{M} \text{ and } a \in C. \quad (12)$$

Proof. Since A is finite and $p(a) > 0$ for all a , choose $\alpha > 0$ so small that

$$\alpha p(a) \in (0, r) \quad \text{for all } a \in A.$$

Because g is strictly monotone, it is injective and has a well-defined inverse on its range. Define

$$u(a) = g^{-1}(\alpha p(a)).$$

Then

$$g(u(a)) = \alpha p(a).$$

For any context C ,

$$\sum_{b \in C} g(u(b)) = \alpha \sum_{b \in C} p(b) = \alpha,$$

because p is an admissible weight. Therefore, for $a \in C$,

$$\frac{g(u(a))}{\sum_{b \in C} g(u(b))} = \frac{\alpha p(a)}{\alpha} = p(a).$$

The resulting probabilities are single-valued on atoms. $\square$

The construction is explicit: given p and g , the scores are

$$u(a) = g^{-1}(\alpha p(a))$$

for any sufficiently small $\alpha > 0$ such that $\alpha p(a)$ lies in the range of g for every atom a .

The theorem is deliberately deflationary. If the scores are completely unrestricted, generalized softmax has no structural content beyond positivity, local normalization, and the imposed gluing condition. Its explanatory force begins only when the scores are constrained independently of the target probabilities, for example by atom-only scores, low-dimensional parametric forms, smoothness conditions across contexts, or a mechanistic model of how utilities or evidential strengths are generated.

Remark IV.3. The hypothesis that the range of g contains an interval $(0, r)$ is only a convenient sufficient condition. For a fixed weight p , it is enough that the range of g contain a scaled copy $\alpha\{p(a) : a \in A\}$ for some $\alpha > 0$. The interval condition makes this automatic for all strictly positive weights on a finite event structure.

Remark IV.4 (Gauge freedom). The generalized softmax representation is not unique. Multiplying all positive coordinates $q(a) = g(u(a))$ in a connected component of the pasted logic by a common positive constant leaves all normalized probabilities unchanged. For the exponential link $g(u) = e^{\beta u}$, this corresponds to adding a constant to all scores in that connected component:

$$u(a) \mapsto u(a) + c.$$

For disconnected logics, this gauge freedom is componentwise. Thus generalized softmax gives a many-to-one parametrization of the relative interior of $\mathcal{W}(L)$, not a unique coordinate chart.

Boundary weights require a separate comment because many interesting exotic weights assign zero probability to some atoms.

Proposition IV.5 (Boundary weights). *Let $p \in \mathcal{W}(L)$. If p lies in the closure of the strictly positive part of $\mathcal{W}(L)$, and if 0 lies in the closure of the range of g , then p lies in the closure of the generalized-softmax image.*

Proof. Choose a sequence $p_k \in \mathcal{W}(L)$ with $p_k(a) > 0$ for all atoms a and $p_k \rightarrow p$. By Theorem IV.2, each p_k has a generalized softmax representation. Taking the limit gives p . Atoms with $p(a) = 0$ correspond to coordinates $g(u_k(a)) \rightarrow 0$. For the exponential link this means $u_k(a) \rightarrow -\infty$. $\square$

For the ordinary exponential softmax, exact zeros therefore require extended scores or limiting procedures. Other response maps, such as sparsemax or entmax, can produce exact zeros at finite scores, but they are typically piecewise analytic rather than strictly positive link normalizations of the form (6).

The theorem has a deflationary consequence. Once single-valuedness is imposed, generalized softmax is not an additional probability theory. It is an analytic, many-to-one coordinate system on $\mathcal{W}(L)$, or on its positive part. The classification of a weight as classical, quantum, or exotic is not determined by the link function g , but by the position of the resulting p inside or outside the relevant subsets of $\mathcal{W}(L)$.

V. RELATION TO CAUCHY'S FUNCTIONAL EQUATION

The preceding results should be distinguished from Cauchy-type linearity theorems. The generalized-softmax construction concerns local normalization and cross-context gluing. Once the probabilities glue, one obtains an admissible weight

$$p \in \mathcal{W}(L),$$

that is, a single-valued function on atoms satisfying

$$\sum_{a \in C} p(a) = 1$$

for every context C . Additivity inside a Boolean block is then imposed by defining the probability of a coarse-grained event as the sum of the probabilities of its mutually exclusive atoms.

This is formally related to Cauchy's functional equation. In Gleason-type theorems, one considers additive probability assignments on richer structures, such as projection lattices or effect algebras. The additivity condition has the Cauchy form

$$f(x + y) = f(x) + f(y), \quad (13)$$

with x and y replaced by compatible effects or orthogonal projections. Under positivity, boundedness, continuity, or measurability assumptions, such additive functions are forced to be linear, yielding the usual trace form of the Born rule. Wright and Weigert [9] explicitly exploit this connection to give an alternative proof of Busch's Gleason-type theorem for effects [12, 13].

The softmax result proved here is different. It does not derive linearity of a probability assignment, nor does it derive the Born rule. It shows that, once single-valuedness or no-disturbance is imposed, generalized softmax normalization is merely a parametrization of an admissible weight. Whether that weight is classical, Born-rule realizable, or exotic is a further geometric question about its location in $\mathcal{W}(L)$.

There is, however, another Cauchy-type route by which the ordinary exponential softmax can be singled out from the class of generalized links. Let $q(u) > 0$ be the unnormalized weight associated with a score u . If independent score contributions are required to add, while their unnormalized weights multiply, then

$$q(u + v) = q(u)q(v). \quad (14)$$

This composition law is natural in settings where scores represent additive evidence, utility increments, or energy-like contributions, while the unnormalized weights represent independent multiplicative factors. In statistical mechanics this is the familiar passage from additive energies to Boltzmann factors; in response models it is the assumption that independent score increments multiply odds-like weights. It is not required by the gluing theorem.

Taking logarithms gives

$$\log q(u + v) = \log q(u) + \log q(v), \quad (15)$$

which is Cauchy's equation.

Proposition V.1 (Exponential link from score composition). *Let $q : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ satisfy*

$$q(u + v) = q(u)q(v) \quad \text{for all } u, v \in \mathbb{R}.$$

If q is measurable, or monotone, or bounded above on a measurable set of positive measure, then there exists $\beta \in \mathbb{R}$ such that

$$q(u) = e^{\beta u}.$$

Proof. Define $h(u) = \log q(u)$. Since $q(u) > 0$, this is well defined. Equation (14) gives

$$h(u + v) = h(u) + h(v).$$

Under any of the stated regularity assumptions, $h = \log q$ is a regular solution of Cauchy's equation: measurability and monotonicity are inherited from q , while boundedness above of q on a measurable set of positive measure

implies boundedness above of h there. Hence the standard regularity theorem for Cauchy's equation gives [14]

$$h(u) = \beta u$$

for some $\beta \in \mathbb{R}$. Hence

$$q(u) = e^{h(u)} = e^{\beta u}.$$

□

A complementary way to single out the exponential link is the maximum-entropy characterization of softmax [15]. If a context C is fixed and one maximizes Shannon entropy over probability assignments $(p_a)_{a \in C}$, subject to normalization and a linear constraint on the expected score,

$$\sum_{a \in C} p_a = 1, \quad \sum_{a \in C} p_a u(a) = \bar{u},$$

then the resulting distribution has the ordinary exponential form

$$p_a = \frac{e^{\beta u(a)}}{\sum_{b \in C} e^{\beta u(b)}}.$$

From this maximum-entropy perspective, the exponential softmax is the canonical context-wise distribution compatible with a fixed expected score. The gluing condition studied in the present paper then appears as an additional cross-context constraint: when the same atom occurs in two contexts, the two locally maximum-entropy distributions must assign it the same probability. Thus the maximum-entropy derivation selects the exponential form inside a single context, whereas single-valuedness or no-disturbance determines whether the resulting context-wise distributions assemble into a global weight on the pasted logic. The Cauchy/composition argument above is dual in spirit: instead of deriving exponentials from an entropy variational principle, it derives them from multiplicativity of unnormalized weights under additive score composition.

Thus Cauchy's equation can justify the exponential link under an additional compositional axiom on scores. The gluing theorem itself, however, does not depend on this axiom and applies equally to any positive link function with suitable range.

The conceptual separation is therefore as follows. Cauchy/Gleason arguments concern the additivity and linear extension of probability weights on sufficiently rich additive domains. The present softmax argument concerns the prior compatibility question of whether locally normalized context-wise response probabilities define a single admissible weight at all. Finally, the special status of the ordinary exponential link arises only if one imposes a separate composition law connecting additive scores to multiplicative unnormalized weights.

VI. CLASSICAL, QUANTUM, AND EXOTIC REGIONS

The admissible-weight polytope $\mathcal{W}(L)$ contains every single-valued probability assignment compatible with the context-wise normalization constraints. The classical polytope $\mathcal{C}(L)$ is generally smaller. It consists of convex mixtures of dispersion-free assignments. In a partition model with a specified latent state space S , the intended classical model may be the convex hull of the point-evaluation states induced by S .

A weight

$$p \in \mathcal{W}(L) \setminus \mathcal{C}(L)$$

is nonclassical in the sense that it cannot be written as a classical mixture of deterministic two-valued states. Such weights are often called exotic in the literature on generalized urns and quantum logics [1, 2]. They are normalized and single-valued as weights on the logic, but they need not admit an interpretation as ignorance over pre-existing values.

A further distinction concerns quantum realizability. A weight may or may not be representable by a Born rule in a Hilbert-space model. If a weight belongs to

$$\mathcal{W}(L) \setminus \mathcal{Q}(L),$$

then it is beyond the specified quantum set. This claim is representation-dependent: one must specify the Hilbert-space dimension, rank of projectors, and exclusivity convention. In odd-cycle scenarios with the usual Lovász-theta convention, the quantum maximum of the cyclic sum is given by Eq. (19) below, whereas the admissible half-weight gives $n/2$.

Generalized softmax coordinates do not change these distinctions. The same link function can represent a classical weight, a quantum-realizable nonclassical weight, or an exotic beyond-quantum weight, provided the target weight lies in the appropriate part of $\mathcal{W}(L)$. Thus one should distinguish carefully between:

1. *Local normalization*: probabilities sum to one inside each context.
2. *Single-valuedness/no-disturbance*: shared atoms receive the same probability across contexts.
3. *Cauchy–Gleason linearity*: on richer domains, Cauchy/Gleason-type assumptions may force additive probability assignments to be linear.
4. *Realizability*: the resulting weight is classical, Born-rule realizable, exotic, or beyond the chosen quantum set.

Softmax and generalized softmax rules address the first item automatically. They address the second only if cross-context constraints are imposed. They do not decide the third or fourth.

Thus admissible probabilities on a pasted logic can go beyond classical, and in some cases beyond quantum, bounds. The odd-cycle half-weights provide the simplest example: they are normalized and single-valued on every context, hence belong to $\mathcal{W}(L)$, but they lie outside the classical convex hull; for odd cycles of length at least five they also exceed the usual Lovász/KCBS quantum bound. This beyond-classical or beyond-quantum character is not produced by softmax normalization itself. It is a property of the event structure and of the selected admissible weight. Softmax or generalized softmax coordinates merely provide a parametrization—or, for boundary weights such as the half-weight, a limiting parametrization—of that weight.

VII. ODD CYCLES AND EXOTIC HALF-WEIGHTS

Odd cyclic logics provide the simplest examples in which admissible weights exceed the classical convex hull. Even cycles admit an alternating two-coloring of their cyclic atoms, and their independence number is $n/2$. The half-weight constructed below therefore saturates but does not exceed the corresponding classical bound for even n , and no exotic behavior arises in the present sense. The interesting case is odd n .

Let $n \geq 3$ be odd. Consider the cyclic pasted logic with atoms

$$a_1, \dots, a_n, \quad x_1, \dots, x_n,$$

and contexts

$$C_i = \{a_i, a_{i+1}, x_i\}, \quad i = 1, \dots, n, \quad (16)$$

where indices are understood modulo n . The atoms a_i are the cyclic intertwining atoms; the atoms x_i are context-specific.

The assignment

$$p(a_i) = \frac{1}{2}, \quad p(x_i) = 0 \quad (i = 1, \dots, n) \quad (17)$$

is an admissible weight, since

$$p(a_i) + p(a_{i+1}) + p(x_i) = \frac{1}{2} + \frac{1}{2} + 0 = 1$$

for every context C_i .

It is not a classical mixture of two-valued states. In any dispersion-free assignment, no two adjacent cyclic atoms can both receive value one. Hence the number of cyclic atoms assigned value one is at most the independence number of the odd cycle,

$$\alpha(C_n) = \frac{n-1}{2}.$$

Therefore every classical mixture satisfies

$$\sum_{i=1}^n p(a_i) \leq \frac{n-1}{2}. \quad (18)$$

The half-weight (17) gives

$$\sum_{i=1}^n p(a_i) = \frac{n}{2},$$

and therefore violates the classical bound by $1/2$.

For the usual exclusivity-graph quantum model on an odd cycle, the Lovász-theta value is [6]

$$\vartheta(C_n) = \frac{n \cos(\pi/n)}{1 + \cos(\pi/n)}. \quad (19)$$

For odd $n \geq 5$,

$$\frac{n}{2} > \frac{n \cos(\pi/n)}{1 + \cos(\pi/n)}.$$

Indeed, since $0 < \pi/n < \pi/2$, one has $\cos(\pi/n) < 1$, and hence

$$1 + \cos(\pi/n) > 2 \cos(\pi/n),$$

which is exactly the displayed inequality after multiplication by $n/[2(1 + \cos(\pi/n))]$. Thus the half-weight also lies beyond the usual quantum set for these odd-cycle exclusivity scenarios. For $n = 5$, Eq. (19) gives $\sqrt{5}$, recovering the KCBS value.

a. Generalized-softmax path between midpoint states and half-weights. The softmax construction can be stated in a way that makes the inessential role of the exponential link explicit. Let g be any strictly positive link function and choose a symmetric global score assignment

$$u(a_i) = u_0, \quad u(x_i) = t \quad (i = 1, \dots, n).$$

Then each context $C_i = \{a_i, a_{i+1}, x_i\}$ has normalizer

$$Z_i = 2g(u_0) + g(t),$$

and therefore

$$p_t(a_i) = p_t(a_{i+1}) = \frac{g(u_0)}{2g(u_0) + g(t)},$$

$$p_t(x_i) = \frac{g(t)}{2g(u_0) + g(t)}.$$

Equivalently, writing

$$r(t) = \frac{g(t)}{g(u_0)}$$

gives the link-independent form

$$p_t(a_i) = p_t(a_{i+1}) = \frac{1}{2 + r(t)}, \quad p_t(x_i) = \frac{r(t)}{2 + r(t)}.$$

This family is single-valued on shared atoms: every cyclic atom a_i has the same probability in the two contexts in which it occurs.

The parameter $r(t)$ interpolates between three instructive regimes. For $r(t) \rightarrow \infty$, one obtains the deterministic midpoint assignment

$$p_t(x_i) \rightarrow 1, \quad p_t(a_i) \rightarrow 0.$$

This is a valid two-valued weight on the pasted logic. What fails on an odd cycle is a stronger symmetric coloring picture in which, after fixing all midpoint atoms, the two remaining labels alternate consistently around the cycle. For $r(t) = 1$, all three atoms in each context have equal positive coordinate and

$$p_t(a_i) = p_t(a_{i+1}) = p_t(x_i) = \frac{1}{3}.$$

For $r(t) \rightarrow 0$, the midpoint atoms are suppressed and one obtains the exotic boundary half-weight

$$p_t(a_i) \rightarrow \frac{1}{2}, \quad p_t(x_i) \rightarrow 0,$$

which is Eq. (17). Thus generalized softmax supplies a smooth fractional path through the same odd-cycle obstruction: the failed integral alternation of colors is replaced, at the opposite boundary, by the symmetric fractional assignment $1/2, 1/2, 0$ in every context.

For the ordinary exponential link $g(u) = e^{\beta u}$, one may set $u_0 = 0$. Then

$$r(t) = e^{\beta t},$$

so the three regimes correspond to $t \rightarrow +\infty$, $t = 0$, and $t \rightarrow -\infty$, respectively, assuming $\beta > 0$. Equivalently, setting $t = -M$ and sending $M \rightarrow \infty$ gives the limiting construction

$$p_t(a_i) \rightarrow \frac{1}{2}, \quad p_t(x_i) \rightarrow 0.$$

For finite real-valued scores the exponential softmax remains strictly positive and therefore approximates, but does not exactly attain, this boundary weight. For a general positive link g , the half-weight is approached whenever

$$r(t) = \frac{g(t)}{g(u_0)} \rightarrow 0,$$

or, equivalently, whenever the positive coordinate assigned to the context-specific atom x_i is suppressed relative to the cyclic atoms. This requires 0 to lie in the closure of the attainable coordinate ratios.

Table I records the explicit probability values for the three distinguished assignments in the pentagon case, and Fig. 1 shows the corresponding logic in the notation used here.

It is worth noting, however, that for every odd $n \geq 3$ the two-valued state

$$p(x_i) = 1, \quad p(a_i) = 0 \quad (i = 1, \dots, n)$$

TABLE I. Explicit probability values for the ten atoms of the pentagon logic in the three distinguished regimes of the path: the midpoint limit $r \rightarrow \infty$, the uniform state $r = 1$, and the half-weight limit $r \rightarrow 0$, where $r = g(t)/g(u_0)$. For the ordinary exponential parametrization with $u_0 = 0$ and $\beta > 0$, these correspond to $t \rightarrow +\infty$, $t = 0$, and $t \rightarrow -\infty$, respectively.

Regime	a_1	x_1	a_2	x_2	a_3	x_3	a_4	x_4	a_5	x_5
Midpoint limit	0	1	0	1	0	1	0	1	0	1
Uniform state	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
Half-weight	$\frac{1}{2}$	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0	$\frac{1}{2}$	0

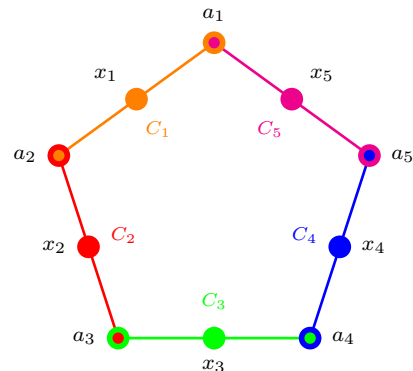


FIG. 1. Pentagon logic in the notation of Sec. VII. The cyclic atoms $a_1, \dots, a_5$ are intertwining atoms, each shared by two adjacent contexts. The atoms $x_1, \dots, x_5$ are context-specific. Each colored broken line represents one context $C_i = \{a_i, a_{i+1}, x_i\}$, with indices understood modulo 5.

is classical as a dispersion-free weight but “exotic” in a different, coloring-theoretic sense: it cannot be refined to a consistent three-coloring of the odd-cycle contexts. Indeed, fixing all midpoint atoms x_i to one color would force the two remaining colors to alternate on the cyclic atoms a_i around an odd cycle, which is impossible [16]. Thus the coloring-theoretic obstruction already appears for the triangle $n = 3$. The pentagon shown in Fig. 1 and Table I is the first odd-cycle case where the classical, Lovász/KCBS, and admissible half-weight bounds are all distinct.

A. Triangle

For $n = 3$, the construction gives Wright’s triangle logic. The half-weight has

$$\sum_{i=1}^3 p(a_i) = \frac{3}{2},$$

whereas the classical bound is

$$\alpha(C_3) = 1.$$

Thus the triangle already exhibits the basic phenomenon: an admissible single-valued weight can lie outside the classical convex hull. The graph-theoretic quantum comparison at $n = 3$, however, is degenerate. The three-cycle coincides with the complete graph K_3 , and it is not a KCBS-type scenario in the usual sense. We therefore use the triangle only as the minimal example of an admissible but nonclassical weight. The pentagon $n = 5$, shown in Fig. 1, is the first nontrivial case in which the classical bound, the quantum Lovász/KCBS bound, and the admissible half-weight are all distinct.

B. Pentagon

For $n = 5$, the construction gives the pentagon/pentagram logic of Fig. 1. The ten atoms are

$$a_1, \dots, a_5, \quad x_1, \dots, x_5,$$

with contexts

$$C_i = \{a_i, a_{i+1}, x_i\},$$

where indices are understood modulo 5. The three distinguished assignments along the softmax path are listed explicitly in Table I.

For the half-weight,

$$p(a_i) = \frac{1}{2}, \quad p(x_i) = 0,$$

one obtains

$$\sum_{i=1}^5 p(a_i) = \frac{5}{2}.$$

This exceeds the classical bound

$$\alpha(C_5) = 2$$

and also the usual quantum/KCBS bound

$$\vartheta(C_5) = \sqrt{5}.$$

Thus the pentagon half-weight is admissible, nonclassical, and beyond the standard quantum five-cycle model.

Along the generalized-softmax path, the cyclic weight is

$$\sum_{i=1}^5 p_t(a_i) = \frac{5}{2 + r(t)}, \quad r(t) = \frac{g(t)}{g(u_0)}.$$

It exceeds the classical bound 2 precisely when

$$\frac{5}{2 + r(t)} > 2,$$

that is,

$$r(t) < \frac{1}{2}.$$

It exceeds the usual KCBS/Lovász value $\sqrt{5}$ precisely when

$$\frac{5}{2 + r(t)} > \sqrt{5},$$

or equivalently

$$r(t) < \sqrt{5} - 2.$$

Thus finite generalized-softmax coordinates already enter the nonclassical and beyond-quantum regions once the midpoint atoms x_i are sufficiently suppressed relative to the cyclic atoms a_i . The exact half-weight $5/2$, however, is reached only in the boundary limit $r(t) \rightarrow 0$. For the ordinary exponential link, this condition is $e^{\beta t} \rightarrow 0$, i.e. $t \rightarrow -\infty$ for $\beta > 0$.

The softmax representation therefore does not explain the beyond-quantum character of the weight. It only parametrizes, in the closure of the positive generalized-softmax image, a point already present in the admissible-weight polytope $\mathcal{W}(L)$.

VIII. EMPIRICAL AND MODELLING IMPLICATIONS

In cognitive and social-science applications, a score $u_C(a)$ may represent utility, salience, evidential strength, diagnostic support, response tendency, or contextual attractiveness. Equation (6) then turns such scores into a probability distribution over the outcomes available in context C .

The preceding analysis shows that this step should not be confused with a global probabilistic model. A context-wise softmax may fit response frequencies in each context separately while failing to assign a single probability to atoms shared by multiple contexts. In such a case the model describes contextual response formation, not a single-valued weight on a pasted logic.

For empirical applications, the analysis suggests a three-stage procedure. First, estimate the response probabilities separately in each context. Second, test whether shared atoms have equal probabilities across the contexts in which they occur, within sampling error. In a between-subjects design, these estimates come from different samples, so a failure of equality may reflect genuine disturbance, selection effects, or a mis-specified exclusivity structure rather than a single identifiable mechanism. Third, if the single-valuedness condition holds to the required statistical precision, reconstruct the global weight $p \in \mathcal{W}(L)$ and test its location relative to the classical polytope $\mathcal{C}(L)$ and, where appropriate, a specified quantum set $\mathcal{Q}(L)$. Membership in $\mathcal{C}(L)$ is a linear-programming question; for small logics it can also be tested directly by enumerating the two-valued states. Membership in common graph-theoretic relaxations of $\mathcal{Q}(L)$ is typically treated by semidefinite programming or by known bounds such as Eq. (19).

Generalized softmax coordinates may be useful for parametrization and estimation, but they do not replace these structural tests. A successful softmax fit inside each context does not tell us whether the observed probabilities are classical, quantum-like, exotic, or not even globally compatible. The decisive issue is not local normalization but the geometry of the global weight.

The representation theorem also shows where genuine modelling assumptions enter. If the scores are completely unrestricted, then a generalized softmax representation has little explanatory content: any strictly positive admissible weight can be represented. A substantive softmax model must therefore constrain the scores independently of the target probabilities. Examples include atom-only scores $u_C(a) = u(a)$, low-dimensional parametric forms

$$u_C(a) = \theta \cdot \phi(a, C),$$

regularity conditions across contexts, or mechanistic assumptions about how utilities, saliences, or evidential strengths are formed. Only such restrictions turn generalized softmax from a reparametrization into an explanatory model.

This hierarchy is particularly relevant to quantum cognition [17, 18]. Context-wise response models are often descriptively flexible, but flexibility is not the same as a structural explanation. The relevant structural questions are whether the fitted probabilities glue, whether the glued weight is classical, whether it is Born-rule realizable, whether a Cauchy/Gleason-type linearity theorem applies on a richer domain, and whether the weight is exotic relative to the chosen event structure.

IX. DISCUSSION

The main result can be summarized as follows:

single-valued generalized softmax
= admissible weight in link-function coordinates.

This equivalence is mathematically elementary, but it prevents a common conflation. Softmax normalization is sometimes treated as a probabilistic principle. In a non-Boolean event structure, however, it is only a local normalization scheme unless cross-context gluing conditions are added. Once those conditions are added, softmax no longer defines a new model class; it simply parametrizes the admissible weights that were already present.

This observation also clarifies the role of link functions. The exponential is not essential for the gluing theorem. Any positive link with a suitable range can be used to coordinatize the positive part of the weight polytope. Changing the link changes the score scale, response sensitivity, and statistical convenience, but not the underlying distinction between classical, quantum, and exotic weights. The exponential link becomes distinguished

only under an additional Cauchy-type composition principle: additive score contributions must correspond to multiplicative unnormalized weights.

The exotic examples are especially instructive. The odd-cycle half-weights are single-valued in the operational sense: each atom has one value and each context is normalized. They are nevertheless not classical mixtures of two-valued states. For odd cycles of length at least five, the same assignments exceed the usual Lovász/KCBS quantum bounds. A limiting softmax can represent these weights, but it does not explain them. Their existence is a property of the pasted event structure.

One should therefore be cautious with phrases such as “softmax beyond quantum.” The more precise statement is: generalized softmax coordinates can represent admissible weights that are beyond the classical or quantum sets. Boundary exotic weights with zero components are represented only as limits or, equivalently, as points in the closure of the positive generalized-softmax image. The beyondness is not due to the normalizing function; it is due to the selected point in $\mathcal{W}(L)$.

This distinction echoes the classic Bell–Peres intuition regarding contextuality. If one insists on assigning outcomes to unperformed measurements, Bell-type violations force the value assigned to an observable to depend on the context in which it is considered. Peres expressed this by replacing a putative setting-independent result $r_{j\alpha}$ by an explicitly context-dependent result $r_{j\alpha}(\beta)$, or more generally $r_{j\alpha}(\beta, \gamma, \dots)$, where the additional arguments refer to the remaining measurement context [19, p. 747]. Similarly, in an explicitly contextual Clauser–Horne–Shimony–Holt (CHSH) notation one may write $E(x, y)$ as $E(x_y, y_x)$, with contextuality manifesting itself through $x_y \neq x_{y'}$ when $y \neq y'$ [20].

In the present framework, a free context-wise softmax without the gluing condition is a probabilistic analogue of this explicit context-dependence. A single nominal atom a is replaced by context-indexed copies $a_C, a_{C'}, \dots$, and the probabilities assigned to these copies need not agree. In contrast, an admissible weight on a pasted event structure imposes the strict identification of shared atoms: if $a \in C \cap C'$, then a carries one probability $p(a)$, not two.

This distinction changes the mathematical problem. In a strictly pasted event structure, shared atoms are the points at which Boolean blocks are glued together. They are responsible for the global constraints that make Bell-, KCBS-, and Kochen–Specker-type obstructions possible. If the shared atoms are split into context-indexed copies, the original pasted structure is replaced by a disjoint family of context-wise Boolean algebras. On that disjoint structure alone there is no logical obstruction to forming a global joint distribution over the context-indexed variables; one may, for example, take a product of the context-wise distributions. What has been removed is not contextuality as an empirical phenomenon, but the strict logical identification of atoms across contexts.

The Contextuality-by-Default (CbD) framework [21, 22] starts precisely from this context-indexed descrip-

tion. Random variables measuring the same content in different contexts are not identified at the outset; they are treated as distinct variables belonging to different bunches. Empirical disturbance, or inconsistent connectedness, is therefore built into the formalism. CbD then reintroduces a probabilistic analogue of gluing by asking whether these context-indexed variables can be coupled so that same-content variables agree as often as their individual distributions allow. A system is contextual, in the CbD sense, if no global coupling can satisfy all these maximal-connection requirements simultaneously. This residual system-wide obstruction is what CbD calls covert context-dependence [22].

Thus the two approaches address different regimes. CbD provides a probabilistic methodology for systems in which context-indexing and empirical disturbance are taken as primitive. The present paper instead isolates the strictly glued regime of pasted event structures. In that regime, shared atoms are identified from the start, and an admissible weight is a single-valued function on those atoms. The main result then says that, once this strict single-valuedness is imposed, generalized softmax normalization contributes no new probabilistic mechanics: it is only a coordinate parametrization of the admissible-weight polytope $\mathcal{W}(L)$. Whether the resulting weight is classical, quantum-realizable, exotic, or beyond a specified quantum set is determined by the geometry of the pasted event structure, not by the softmax link.

A natural next step is not to ask whether generalized softmax can represent exotic weights—it can, at least in the closure—but which constrained score families can do so. Atom-only scores $u_C(a) = u(a)$, low-dimensional forms $u_C(a) = \theta \cdot \phi(a, C)$, locality restrictions on ϕ , or smoothness assumptions across neighboring contexts may cut out proper subregions of $\mathcal{W}(L)$. Characterizing these subregions, and comparing them with $\mathcal{C}(L)$, $\mathcal{Q}(L)$, and the exotic part of $\mathcal{W}(L)$, is the substantive modelling problem left open by the present deflationary result.

Several extensions suggest themselves. First, one may study sparse or thresholded response maps that produce zero weights at finite scores. Second, one may impose additional locality, dimensionality, or regularity constraints on the score function u , thereby obtaining proper subfamilies of $\mathcal{W}(L)$. Third, one may investigate statistical estimation procedures in which generalized softmax coordinates are used for constrained inference over $\mathcal{C}(L)$, $\mathcal{Q}(L)$, or $\mathcal{W}(L)$. Fourth, one may ask when Cauchy/Gleason-type extension theorems apply to weights obtained from empirical response models, thereby connecting finite pasted logics with richer effect-algebraic or Hilbert-space structures. These directions would turn the present geometric observation into a practical modeling framework.

Appendix A records a Lean 4/mathlib formalization of the finite representation, gluing, and pentagon half-weight claims. The verified artifacts are included verbatim so that the computational proof certificate can be inspected independently of the surrounding exposition.

X. CONCLUSION

Generalized softmax rules transform scores into normalized probabilities within a context. In non-Boolean event structures with overlapping contexts, this is not enough to define a probability weight. Shared atoms must receive identical probabilities wherever they occur. If this no-disturbance or single-valuedness condition is not imposed, one has a family of context-wise response distributions, not a weight on the pasted logic. If it is imposed, the generalized softmax rule collapses to a coordinate representation of an admissible weight.

Under mild assumptions on the positive link function, every strictly positive admissible weight can be represented in this way. Boundary weights, including the exotic half-weights on odd cycles, are obtained as limits when zero lies in the closure of the link’s range. Therefore single-valued generalized softmax is not a new probability theory. It is a parametrization of $\mathcal{W}(L)$. The interesting distinctions are instead geometric: whether the resulting weight is classical, quantum-realizable, exotic, or beyond the specified quantum set.

This result is adjacent to, but distinct from, Cauchy/Gleason linearity theorems. Cauchy-type additivity arguments can force linear/Born-rule probability assignments on sufficiently rich additive domains. The present theorem instead concerns the gluing of locally normalized response probabilities into a single admissible weight. Cauchy’s equation reappears at the level of the link function only if one imposes the additional axiom that additive scores give multiplicative unnormalized weights, in which case the ordinary exponential softmax is singled out.

Odd cyclic logics provide minimal examples. They support half-weights that are normalized and single-valued but not classical; for odd cycles of length at least five, these weights exceed the usual Lovász/KCBS quantum bound. They are obtained as limiting softmax assignments, but their exoticity belongs to the event structure and the admissible-weight polytope, not to the exponential function.

The resulting lesson is simple: local normalization, single-valuedness/no-disturbance, Cauchy/Gleason linearity, and physical or cognitive realizability are distinct. Softmax addresses the first. The pasted logic imposes the second. Cauchy-type arguments may address the third on richer domains. Classical, quantum, and exotic probability theories concern the fourth.

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Appendix A: Lean verification artifacts

This appendix records the Lean 4 formalization used to verify the finite claims about gluing, generalized-softmax coordinates, and the pentagon half-weight. The project was checked with Lean v4.30.0 and mathlib v4.30.0; the command `lake build` completed successfully, and the source contains no `sorry`, `admit`, or `axiom`.

The formalization separates three layers. First, Appendix A 2 defines finite event structures, admissible weights, positive coordinates, and the representation theorem for scaled link-function preimages. Second, Appendix A 3 formalizes local context-wise distributions, single-valuedness, the normalizer-ratio calculation, and the theorem that single-valued local distributions glue to a global admissible weight. Third, Appendix A 4 formalizes the pentagon example: the half-weight is admissible, every two-valued state and every finite classical mixture obeys the cyclic bound 2, the half-weight has cyclic sum $5/2$, and $5/2 > \sqrt{5}$. The same file also verifies the positive softmax-coordinate path and its boundary endpoint at the half-weight.

TABLE II. Lean files included in the verification appendix.

File	Main content
<code>lean-toolchain</code>	Lean compiler pin
<code>lakefile.lean</code>	mathlib dependency pin
<code>SoftmaxVerification.lean</code>	project import root
<code>Core.lean</code>	event structures and representation theorem
<code>Gluing.lean</code>	gluing and softmax-to-weight theorem
<code>Pentagon.lean</code>	pentagon half-weight and bounds

1. Project files

a. lean-toolchain. Lean compiler pin.

```
1 leanprover/lean4:v4.30.0
```

b. lakefile.lean. Lake project file and mathlib dependency.

```
1 import Lake
2 open Lake DSL
3
4 package SoftmaxVerification where
5
6 require mathlib from git
7   "https://github.com/leanprover-community/mathlib4.git" @ "v4.30.0"
8
9 @[default_target]
10 lean_lib SoftmaxVerification where
```

c. SoftmaxVerification.lean. Root Lean import file.

```
1 import SoftmaxVerification.Core
2 import SoftmaxVerification.Gluing
3 import SoftmaxVerification.Pentagon
```

2. Core representation theorem

a. SoftmaxVerification/Core.lean. Finite event structures, admissible weights, positive coordinates, and the link-function representation theorem.

```
1 import Mathlib
2
3 open scoped BigOperators
4
5 namespace SoftmaxVerification
6
```

```

7  /--
8  A finite event structure in the sense used in the softmax paper.
9
10 `atoms` is the ambient finite set of atoms. Each context is itself a finite
11 set of atoms, and `context_subset` records that contexts do not contain
12 anything outside the ambient atom set.
13 -/
14 structure EventStructure (Atom : Type*) [DecidableEq Atom] where
15   atoms : Finset Atom
16   contexts : Finset (Finset Atom)
17   context_subset : forall C, C ∈ contexts -> C ⊆ atoms
18
19 /--
20 An admissible weight on an event structure: nonnegative on atoms and
21 normalized to one on every context.
22 -/
23 structure IsWeight Atom : Type* [DecidableEq Atom]
24   (L : EventStructure Atom) (p : Atom -> Real) : Prop where
25   nonneg : forall a, a ∈ L.atoms -> 0 ≤ p a
26   normalized : forall C, C ∈ L.contexts -> Finset.sum C p = 1
27
28 /-- Strict positivity on all atoms of the event structure. -/
29 def StrictlyPositive Atom : Type* [DecidableEq Atom]
30   (L : EventStructure Atom) (p : Atom -> Real) : Prop :=
31   forall a, a ∈ L.atoms -> 0 < p a
32
33 theorem strictlyPositive_nonneg Atom : Type* [DecidableEq Atom]
34   L : EventStructure Atom p : Atom -> Real
35   (hpos : StrictlyPositive L p) :
36   forall a, a ∈ L.atoms -> 0 ≤ p a := by
37   intro a ha
38   exact le_of_lt (hpos a ha)
39
40 /-- The normalizing denominator for positive coordinates on a context. -/
41 def denom Atom : Type* (q : Atom -> Real) (C : Finset Atom) : Real :=
42   Finset.sum C q
43
44 /-- Context-wise normalization of coordinates. -/
45 noncomputable def normalizedCoordinate Atom : Type*
46   (q : Atom -> Real) (C : Finset Atom) (a : Atom) : Real :=
47   q a / denom q C
48
49 /--
50 `q` represents the admissible weight `p` by normalized coordinates if the
51 coordinates are positive on atoms and context-wise normalization recovers `p`.
52 -/
53 def Represents Atom : Type* [DecidableEq Atom]
54   (L : EventStructure Atom) (p q : Atom -> Real) : Prop :=
55   (forall a, a ∈ L.atoms -> 0 < q a) ∧
56   forall C, C ∈ L.contexts -> forall a, a ∈ C ->
57     normalizedCoordinate q C a = p a
58
59 /--
60 Coordinate form of the paper's representation theorem.
61
62 Every strictly positive admissible weight represents itself: take the
63 unnormalized coordinate of an atom to be its weight.
64 -/
65 theorem positive_weight_represents_itself Atom : Type* [DecidableEq Atom]
66   L : EventStructure Atom p : Atom -> Real
67   (hp : IsWeight L p) (hpos : StrictlyPositive L p) :
68   Represents L p p := by
69   constructor
70   · exact hpos
71   · intro C hC a ha
72     unfold normalizedCoordinate denom
73     rw [hp.normalized C hC]
74     simp
75
76 /--
77 The same representation after multiplying all coordinates by a positive
78 constant. This is the basic gauge freedom of normalized coordinates.
79 -/
80 theorem positive_weight_scaled_coordinates Atom : Type* [DecidableEq Atom]
81   L : EventStructure Atom p : Atom -> Real
82   (hp : IsWeight L p) (hpos : StrictlyPositive L p)
83   alpha : Real (halpha : 0 < alpha) :
84   Represents L p (fun a => alpha * p a) := by
85   constructor
86   · intro a ha
87     exact mul_pos halpha (hpos a ha)
88   · intro C hC a ha
89     unfold normalizedCoordinate denom
90     have hsum : Finset.sum C (fun b => alpha * p b) = alpha := by
91       calc
92         Finset.sum C (fun b => alpha * p b) = alpha * Finset.sum C p := by

```



```

93     rw [Finset.mul_sum]
94     _ = alpha * 1 := by
95     rw [hp.normalized C hC]
96     _ = alpha := by
97     ring
98   rw [hsum]
99   field_simp [ne_of_gt halpha]
100
101 /--
102 Link-function version of the representation theorem.
103
104 Instead of formalizing an inverse function immediately, this theorem states the
105 exact range condition used in the paper: if every scaled value `alpha * p a`
106 has a preimage under the link `g`, then choosing those preimages as scores
107 gives the desired generalized-softmax representation.
108 -/
109 theorem link_representation_of_scaled_range
110   (Atom Score : Type* [DecidableEq Atom] [Nonempty Score])
111   (L : EventStructure Atom p : Atom -> Real)
112   (g : Score -> Real)
113   (hp : IsWeight L p)
114   (alpha : Real (halpha : 0 < alpha))
115   (hpre : forall a, a ∈ L.atoms -> exists u : Score, g u = alpha * p a) :
116   exists u : Atom -> Score,
117     forall C, C ∈ L.contexts -> forall a, a ∈ C ->
118       normalizedCoordinate (fun b => g (u b)) C a = p a := by
119   classical
120   let u : Atom -> Score := fun a =>
121     if ha : a ∈ L.atoms then
122       Classical.choose (hpre a ha)
123     else
124       Classical.choice (inferInstance : Nonempty Score)
125   refine ⟨u, ?_⟩
126   intro C hC a ha
127   have hCsub : C ⊆ L.atoms := L.context_subset C hC
128   have hu : forall b, b ∈ C -> g (u b) = alpha * p b := by
129     intro b hb
130     have hb_atoms : b ∈ L.atoms := hCsub hb
131     dsimp [u]
132     rw [dif_pos hb_atoms]
133     exact Classical.choose_spec (hpre b hb_atoms)
134   unfold normalizedCoordinate denom
135   have hsum : Finset.sum C (fun b => g (u b)) = alpha := by
136     calc
137       Finset.sum C (fun b => g (u b)) = Finset.sum C (fun b => alpha * p b) := by
138         apply Finset.sum_congr rfl
139         intro b hb
140         exact hu b hb
141       _ = alpha * Finset.sum C p := by
142         rw [Finset.mul_sum]
143       _ = alpha * 1 := by
144         rw [hp.normalized C hC]
145       _ = alpha := by
146         ring
147   change g (u a) / Finset.sum C (fun b => g (u b)) = p a
148   rw [hu a ha, hsum]
149   field_simp [ne_of_gt halpha]
150
151 end SoftmaxVerification

```

3. Gluing of local distributions

a. SoftmaxVerification/Gluing.lean. Single-valued local distributions and their glued admissible weight.

```

1 import SoftmaxVerification.Core
2
3 open scoped BigOperators
4
5 namespace SoftmaxVerification
6
7 /-- A context-indexed family of local probabilities is nonnegative on
8 the atoms of each context. -/
9 def LocalNonnegative Atom : Type* [DecidableEq Atom]
10   (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) : Prop :=
11   forall C, C ∈ L.contexts -> forall a, a ∈ C -> 0 <= P C a
12
13 /-- A context-indexed family of local probabilities is normalized in
14 each context. -/
15 def LocalNormalized Atom : Type* [DecidableEq Atom]
16   (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) : Prop :=
17   forall C, C ∈ L.contexts -> Finset.sum C (P C) = 1

```

```

18
19 /-- Single-valuedness/no-disturbance for context-indexed local
20 probabilities. Shared atoms receive the same value in all contexts in
21 which they occur. -/
22 def SingleValuedOn Atom : Type* [DecidableEq Atom]
23   (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) : Prop :=
24   forall C, C ∈ L.contexts -> forall D, D ∈ L.contexts -> forall a,
25     a ∈ C -> a ∈ D -> P C a = P D a
26
27 /-- Local normalizing denominator for freely context-dependent
28 positive coordinates. -/
29 def localDenom Atom : Type*
30   (q : Finset Atom -> Atom -> Real) (C : Finset Atom) : Real :=
31   Finset.sum C (q C)
32
33 /-- Local softmax normalization for freely context-dependent positive
34 coordinates. -/
35 noncomputable def localCoordinate Atom : Type*
36   (q : Finset Atom -> Atom -> Real) (C : Finset Atom) (a : Atom) : Real :=
37   q C a / localDenom q C
38
39 /-- The paper's gluing condition, stated as single-valuedness of the
40 locally normalized coordinates. -/
41 theorem gluing_condition_for_free_scores Atom : Type* [DecidableEq Atom]
42   L : EventStructure Atom q : Finset Atom -> Atom -> Real :
43   SingleValuedOn L (localCoordinate q) <->
44   forall C, C ∈ L.contexts -> forall D, D ∈ L.contexts -> forall a,
45     a ∈ C -> a ∈ D ->
46       localCoordinate q C a = localCoordinate q D a := by
47   rfl
48
49 /-- Algebraic form of the normalizer-ratio condition for one shared
50 atom. This is the scalar calculation behind Eq. (4.3) in the paper. -/
51 theorem normalized_eq_iff_ratio
52   qC qD ZC ZD : Real
53   (hqD : qD ≠ 0) (hZC : ZC ≠ 0) (hZD : ZD ≠ 0) :
54   qC / ZC = qD / ZD <-> ZC / ZD = qC / qD := by
55   constructor
56   · intro h
57     field_simp [hZC, hZD] at h
58     field_simp [hZD, hqD]
59     nlinarith
60   · intro h
61     field_simp [hZD, hqD] at h
62     field_simp [hZC, hZD]
63     nlinarith
64
65 /-- A noncomputable glued global weight: choose one context containing
66 the atom, if such a context exists, and otherwise use zero. The choice
67 is harmless under `SingleValuedOn`. -/
68 noncomputable def gluedWeight Atom : Type* [DecidableEq Atom]
69   (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) :
70   Atom -> Real :=
71   fun a =>
72     if h : exists C, C ∈ L.contexts ∧ a ∈ C then
73       P (Classical.choose h) a
74     else
75       0
76
77 theorem gluedWeight_eq_of_mem Atom : Type* [DecidableEq Atom]
78   L : EventStructure Atom P : Finset Atom -> Atom -> Real
79   (hsingle : SingleValuedOn L P)
80   C : Finset Atom (hC : C ∈ L.contexts) a : Atom (ha : a ∈ C) :
81   gluedWeight L P a = P C a := by
82   classical
83   unfold gluedWeight
84   have hex : exists D, D ∈ L.contexts ∧ a ∈ D := ⟨C, hC, ha⟩
85   rw [dif_pos hex]
86   have hchoose := Classical.choose_spec hex
87   exact hsingle (Classical.choose hex) hchoose.1 C hC a hchoose.2 ha
88
89 /-- Proposition 5.1 in finite form: every single-valued family of
90 locally normalized distributions defines an admissible global weight. -/
91 theorem local_distributions_glue_to_weight Atom : Type* [DecidableEq Atom]
92   L : EventStructure Atom P : Finset Atom -> Atom -> Real
93   (hnonneg : LocalNonnegative L P)
94   (hnorm : LocalNormalized L P)
95   (hsingle : SingleValuedOn L P) :
96   exists p : Atom -> Real,
97     IsWeight L p ∧
98     forall C, C ∈ L.contexts -> forall a, a ∈ C -> p a = P C a := by
99   classical
100   refine ⟨gluedWeight L P, ?_, ?_⟩
101   · constructor
102     · intro a ha
103       unfold gluedWeight

```

```

104   by_cases hex : exists C, C ∈ L.contexts ∧ a ∈ C
105   · rw [dif_pos hex]
106     have hchoose := Classical.choose_spec hex
107     exact hnonneg (Classical.choose hex) hchoose.1 a hchoose.2
108   · rw [dif_neg hex]
109   · intro C hC
110   · calc
111     Finset.sum C (gluedWeight L P) = Finset.sum C (P C) := by
112       apply Finset.sum_congr rfl
113       intro a ha
114       exact gluedWeight_eq_of_mem hsingle hC ha
115     _ = 1 := hnorm C hC
116   · intro C hC a ha
117     exact gluedWeight_eq_of_mem hsingle hC ha
118
119 end SoftmaxVerification

```

4. Pentagon half-weight

a. SoftmaxVerification/Pentagon.lean. Pentagon logic, classical cyclic bound, KCBS numerical comparison, and the positive softmax path.

```

1  import SoftmaxVerification.Gluing
2
3  open scoped BigOperators
4
5  namespace SoftmaxVerification
6
7  /-- The ten atoms of the pentagon logic. -/
8  inductive PentagonAtom where
9    | a0 | a1 | a2 | a3 | a4
10   | x0 | x1 | x2 | x3 | x4
11   deriving DecidableEq, Fintype
12
13  open PentagonAtom
14
15  def pentagonC0 : Finset PentagonAtom := a0, a1, x0
16  def pentagonC1 : Finset PentagonAtom := a1, a2, x1
17  def pentagonC2 : Finset PentagonAtom := a2, a3, x2
18  def pentagonC3 : Finset PentagonAtom := a3, a4, x3
19  def pentagonC4 : Finset PentagonAtom := a4, a0, x4
20
21  def pentagonContexts : Finset (Finset PentagonAtom) :=
22    pentagonC0, pentagonC1, pentagonC2, pentagonC3, pentagonC4
23
24  /-- The pentagon/pentagram event structure from the paper. -/
25  def pentagon : EventStructure PentagonAtom where
26    atoms := Finset.univ
27    contexts := pentagonContexts
28    context_subset := by
29      intro C hC a ha
30      simp
31
32  /-- The cyclic atoms `a_i`. -/
33  def pentagonCyclicAtoms : Finset PentagonAtom := a0, a1, a2, a3, a4
34
35  def pentagonCyclicSum (p : PentagonAtom → Real) : Real :=
36    Finset.sum pentagonCyclicAtoms p
37
38  /-- The pentagon half-weight: cyclic atoms have weight `1/2`, midpoint
39  atoms have weight `0`. -/
40  noncomputable def pentagonHalfWeight : PentagonAtom → Real
41    | a0 | a1 | a2 | a3 | a4 => 1 / 2
42    | x0 | x1 | x2 | x3 | x4 => 0
43
44  theorem pentagon_halfWeight_isWeight : IsWeight pentagon pentagonHalfWeight := by
45    constructor
46    · intro a ha
47      cases a <|> norm_num [pentagonHalfWeight]
48    · intro C hC
49      simp [pentagon, pentagonContexts] at hC
50      rcases hC with hC | hC | hC | hC | hC
51      · subst C
52        simp [pentagonC0, pentagonHalfWeight]
53        norm_num
54      · subst C
55        simp [pentagonC1, pentagonHalfWeight]
56        norm_num
57      · subst C
58        simp [pentagonC2, pentagonHalfWeight]
59        norm_num

```

```

60   · subst C
61     simp [pentagonC3, pentagonHalfWeight]
62     norm_num
63   · subst C
64     simp [pentagonC4, pentagonHalfWeight]
65     norm_num
66
67 theorem pentagon_halfWeight_cyclicSum :
68   pentagonCyclicSum pentagonHalfWeight = 5 / 2 := by
69   simp [pentagonCyclicSum, pentagonCyclicAtoms, pentagonHalfWeight]
70   norm_num
71
72 /-- A two-valued admissible weight. -/
73 def IsTwoValued Atom : Type* [DecidableEq Atom]
74   (L : EventStructure Atom) (v : Atom → Real) : Prop :=
75   IsWeight L v ∧ forall a, a ∈ L.atoms → v a = 0 ∨ v a = 1
76
77 private theorem pentagon_adjacent_sum_le_one
78   v : PentagonAtom → Real (hv : IsTwoValued pentagon v) :
79   v a0 + v a1 ≤ 1 ∧
80   v a1 + v a2 ≤ 1 ∧
81   v a2 + v a3 ≤ 1 ∧
82   v a3 + v a4 ≤ 1 ∧
83   v a4 + v a0 ≤ 1 := by
84   have hx0 : 0 ≤ v x0 := hv.1.nonneg x0 (by simp [pentagon])
85   have hx1 : 0 ≤ v x1 := hv.1.nonneg x1 (by simp [pentagon])
86   have hx2 : 0 ≤ v x2 := hv.1.nonneg x2 (by simp [pentagon])
87   have hx3 : 0 ≤ v x3 := hv.1.nonneg x3 (by simp [pentagon])
88   have hx4 : 0 ≤ v x4 := hv.1.nonneg x4 (by simp [pentagon])
89   have h0 := hv.1.normalized pentagonC0 (by simp [pentagon, pentagonContexts])
90   have h1 := hv.1.normalized pentagonC1 (by simp [pentagon, pentagonContexts])
91   have h2 := hv.1.normalized pentagonC2 (by simp [pentagon, pentagonContexts])
92   have h3 := hv.1.normalized pentagonC3 (by simp [pentagon, pentagonContexts])
93   have h4 := hv.1.normalized pentagonC4 (by simp [pentagon, pentagonContexts])
94   simp [pentagonC0] at h0
95   simp [pentagonC1] at h1
96   simp [pentagonC2] at h2
97   simp [pentagonC3] at h3
98   simp [pentagonC4] at h4
99   constructor
100  · nlinarith
101  constructor
102  · nlinarith
103  constructor
104  · nlinarith
105  constructor
106  · nlinarith
107  · nlinarith
108
109 /-- Every two-valued pentagon weight assigns value one to at most two
110 cyclic atoms. -/
111 theorem pentagon_twoValued_cyclicSum_le_two
112   v : PentagonAtom → Real (hv : IsTwoValued pentagon v) :
113   pentagonCyclicSum v ≤ 2 := by
114   have hadj := pentagon_adjacent_sum_le_one hv
115   rcases hadj with (h01, h12, h23, h34, h40)
116   have ha0 := hv.2 a0 (by simp [pentagon])
117   have ha1 := hv.2 a1 (by simp [pentagon])
118   have ha2 := hv.2 a2 (by simp [pentagon])
119   have ha3 := hv.2 a3 (by simp [pentagon])
120   have ha4 := hv.2 a4 (by simp [pentagon])
121   rcases ha0 with ha0 | ha0 <=>
122   rcases ha1 with ha1 | ha1 <=>
123   rcases ha2 with ha2 | ha2 <=>
124   rcases ha3 with ha3 | ha3 <=>
125   rcases ha4 with ha4 | ha4 <=>
126   simp [pentagonCyclicSum, pentagonCyclicAtoms, ha0, ha1, ha2, ha3, ha4] <=>
127   nlinarith
128
129 /-- A finite classical mixture of two-valued pentagon weights obeys the
130 same cyclic inequality. -/
131 theorem pentagon_classical_mixture_cyclicSum_le_two
132   ι : Type* [Fintype ι]
133   lambda : ι → Real v : ι → PentagonAtom → Real
134   p : PentagonAtom → Real
135   (hlambda_nonneg : forall i, 0 ≤ lambda i)
136   (hlambda_sum : Finset.sum Finset.univ lambda = 1)
137   (hv : forall i, IsTwoValued pentagon (v i))
138   (hp : forall a, p a = Finset.sum Finset.univ (fun i => lambda i * v i a)) :
139   pentagonCyclicSum p ≤ 2 := by
140   have hsum :
141     pentagonCyclicSum p =
142     Finset.sum Finset.univ (fun i => lambda i * pentagonCyclicSum (v i)) := by
143     unfold pentagonCyclicSum
144     calc
145     Finset.sum pentagonCyclicAtoms p =

```

```

146     Finset.sum pentagonCyclicAtoms
147     (fun a => Finset.sum Finset.univ (fun i => lambda i * v i a)) := by
148   apply Finset.sum_congr rfl
149   intro a ha
150   exact hp a
151   _ = Finset.sum Finset.univ
152     (fun i => Finset.sum pentagonCyclicAtoms (fun a => lambda i * v i a)) := by
153   rw [Finset.sum_comm]
154   _ = Finset.sum Finset.univ (fun i => lambda i * Finset.sum pentagonCyclicAtoms (v i)) := by
155   apply Finset.sum_congr rfl
156   intro i hi
157   rw [Finset.mul_sum]
158 rw [hsum]
159 calc
160   Finset.sum Finset.univ (fun i => lambda i * pentagonCyclicSum (v i))
161   <= Finset.sum Finset.univ (fun i => lambda i * 2) := by
162   apply Finset.sum_le_sum
163   intro i hi
164   exact mul_le_mul_of_nonneg_left (pentagon_twoValued_cyclicSum_le_two (hv i))
165   (hlambda_nonneg i)
166   _ = 2 := by
167   rw [← Finset.sum_mul]
168   rw [hlambda_sum]
169   ring
170
171 theorem pentagon_halfWeight_exceeds_classical_bound :
172   2 < pentagonCyclicSum pentagonHalfWeight := by
173   rw [pentagon_halfWeight_cyclicSum]
174   norm_num
175
176 theorem pentagon_halfWeight_exceeds_kcbs_value :
177   Real.sqrt 5 < pentagonCyclicSum pentagonHalfWeight := by
178   rw [pentagon_halfWeight_cyclicSum]
179   rw [Real.sqrt_lt' (by norm_num : 0 < (5 : Real) / 2)]
180   norm_num
181
182 /-- The positive generalized-softmax path used in the paper, written in
183 terms of the coordinate ratio `r`. -/
184 noncomputable def pentagonPathWeight (r : Real) : PentagonAtom -> Real
185 | a0 | a1 | a2 | a3 | a4 => 1 / (2 + r)
186 | x0 | x1 | x2 | x3 | x4 => r / (2 + r)
187
188 theorem pentagon_pathWeight_isWeight r : Real (hr : 0 <= r) :
189   IsWeight pentagon (pentagonPathWeight r) := by
190   have hden_pos : 0 < 2 + r := by linarith
191   constructor
192   · intro a ha
193     cases a <|> dsimp [pentagonPathWeight]
194     · exact div_nonneg zero_le_one hden_pos.le
195     · exact div_nonneg zero_le_one hden_pos.le
196     · exact div_nonneg zero_le_one hden_pos.le
197     · exact div_nonneg zero_le_one hden_pos.le
198     · exact div_nonneg zero_le_one hden_pos.le
199     · exact div_nonneg hr hden_pos.le
200     · exact div_nonneg hr hden_pos.le
201     · exact div_nonneg hr hden_pos.le
202     · exact div_nonneg hr hden_pos.le
203     · exact div_nonneg hr hden_pos.le
204   · intro C hC
205     simp [pentagon, pentagonContexts] at hC
206     rcases hC with hC | hC | hC | hC | hC
207     · subst C
208       simp [pentagonC0, pentagonPathWeight]
209       field_simp [ne_of_gt hden_pos]
210       ring_nf
211     · subst C
212       simp [pentagonC1, pentagonPathWeight]
213       field_simp [ne_of_gt hden_pos]
214       ring_nf
215     · subst C
216       simp [pentagonC2, pentagonPathWeight]
217       field_simp [ne_of_gt hden_pos]
218       ring_nf
219     · subst C
220       simp [pentagonC3, pentagonPathWeight]
221       field_simp [ne_of_gt hden_pos]
222       ring_nf
223     · subst C
224       simp [pentagonC4, pentagonPathWeight]
225       field_simp [ne_of_gt hden_pos]
226       ring_nf
227
228 theorem pentagon_pathWeight_cyclicSum (r : Real) :
229   pentagonCyclicSum (pentagonPathWeight r) = 5 / (2 + r) := by
230   simp [pentagonCyclicSum, pentagonCyclicAtoms, pentagonPathWeight]
231   ring_nf

```



```

232
233 theorem pentagon_path_exceeds_classical_iff r : Real (hr : 0 <= r) :
234   2 < pentagonCyclicSum (pentagonPathWeight r) <-> r < 1 / 2 := by
235   have hden_pos : 0 < 2 + r := by linarith
236   rw [pentagon_pathWeight_cyclicSum]
237   constructor
238   · intro h
239     field_simp [ne_of_gt hden_pos] at h
240     linarith
241   · intro h
242     field_simp [ne_of_gt hden_pos]
243     linarith
244
245 theorem pentagon_path_exceeds_kcbs_of_small r : Real
246   (hr : 0 <= r) (hsmall : r < Real.sqrt 5 - 2) :
247   Real.sqrt 5 < pentagonCyclicSum (pentagonPathWeight r) := by
248   have hden_pos : 0 < 2 + r := by linarith
249   have hsqrt_pos : 0 < Real.sqrt 5 := Real.sqrt_pos.2 (by norm_num)
250   have hlt : 2 + r < Real.sqrt 5 := by linarith
251   have hmul : Real.sqrt 5 * (2 + r) < Real.sqrt 5 * Real.sqrt 5 :=
252     mul_lt_mul_of_pos_left hlt hsqrt_pos
253   have hsq : Real.sqrt 5 * Real.sqrt 5 = 5 := by
254     rw [← pow_two, Real.sq_sqrt]
255     norm_num
256   rw [pentagon_pathWeight_cyclicSum]
257   field_simp [ne_of_gt hden_pos]
258   nlinarith
259
260 theorem pentagon_pathWeight_strictlyPositive r : Real (hr : 0 < r) :
261   StrictlyPositive pentagon (pentagonPathWeight r) := by
262   intro a ha
263   have hden_pos : 0 < 2 + r := by linarith
264   cases a <|> dsimp [pentagonPathWeight]
265   · exact div_pos zero_lt_one hden_pos
266   · exact div_pos zero_lt_one hden_pos
267   · exact div_pos zero_lt_one hden_pos
268   · exact div_pos zero_lt_one hden_pos
269   · exact div_pos zero_lt_one hden_pos
270   · exact div_pos hr hden_pos
271   · exact div_pos hr hden_pos
272   · exact div_pos hr hden_pos
273   · exact div_pos hr hden_pos
274   · exact div_pos hr hden_pos
275
276 theorem pentagon_halfWeight_not_strictlyPositive :
277   ¬ StrictlyPositive pentagon pentagonHalfWeight := by
278   intro hpos
279   have hx := hpos x0 (by simp [pentagon])
280   norm_num [pentagonHalfWeight] at hx
281
282 theorem pentagon_pathWeight_zero_eq_halfWeight :
283   pentagonPathWeight 0 = pentagonHalfWeight := by
284   funext a
285   cases a <|> norm_num [pentagonPathWeight, pentagonHalfWeight]
286
287 end SoftmaxVerification

```

-
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